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ORIGINAL ARTICLE

Relative contributions of anaemia and hypotension to myocardial infarction and renal injury

Post hoc analysis of the POISE-2 trial

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BACKGROUND Hypotension and postoperative anaemia are associated with myocardial and renal injury after noncardiac surgery, but the interaction between them remains unknown.

OBJECTIVES To test the hypothesis that a double-hit of postoperative anaemia and hypotension synergistically worsens a 30-day composite of myocardial infarction (MI) and mortality and acute kidney injury (AKI). Characterising the interaction when hypotension and anaemia occur at same time on myocardial infarction and acute kidney injury.

DESIGN Post hoc analysis of the POISE-2 trial.

SETTING Patients were enrolled between July 2010 and December 2013 at 135 hospitals in 23 countries.

PATIENTS Adults at least 45 years old with known or suspected cardiovascular disease. We excluded patients without available postoperative haemoglobin measurements or hypotension duration records. Exposures were the lowest haemoglobin concentration and the average daily duration of SBP less than 90 mmHg within the first four postoperative days.

MAIN OUTCOME MEASURES The primary outcome was a collapsed composite of nonfatal MI and all-cause mortality during the initial 30 postoperative days; our secondary outcome was AKI.

RESULTS We included 7940 patients. The mean $\pm$ SD lowest postoperative haemoglobin was $10 \pm 2\,g\,dl^{-1}$, and 24% of the patients had SBP less than 90 mmHg with daily duration ranging from 0 to 15 h. Four hundred and nine (5.2%) patients had an infarction or died within 30 postoperative days, and 417 (6.4%) patients developed AKI. Lowest haemoglobin concentrations less than $11\,g\,dl^{-1}$, and duration of SBP less than 90 mmHg was associated with greater hazard of composite outcome of nonfatal MI and all-cause mortality, as well as with AKI. However, we did not find significant multiplicative interactions between haemoglobin splines and hypotension duration on the primary composite or on AKI.

CONCLUSION Postoperative anaemia and hypotension were meaningfully associated with both our primary composite and AKI. However, lack of significant interaction suggests that the effects of hypotension and anaemia are additive rather than multiplicative.

TRIAL REGISTRATION Clinicaltrials.gov: NCT01082874.

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KEY POINTS

- Hypotension and postoperative anaemia are associated with myocardial and renal injury after noncardiac surgery, but the interaction between them remains unknown.
- We characterised the interaction between hypotension and anaemia on myocardial infarction and acute kidney injury.
- Hypotension and anaemia are most likely additive rather than multiplicative on the composite outcome of myocardial infarction and death.
- Clinicians should avoid the combination of anaemia and hypotension in postoperative patients.

Introduction

Myocardial injury/infarction and acute kidney injury (AKI) are common, in postoperative patients.[1,2] Hypotension and anaemia are both associated with myocardial injury, myocardial infarction, acute kidney injury, and all-cause mortality.[1,3–5] Hypotension and anaemia often occur simultaneously. However, the interaction between these two risk factors remains unknown. Possible interactions include antagonistic, additive, multiplicative (synergistic) and permissive (one acts if the other is present but not otherwise). Our goal was, therefore, to characterise the interaction between hypotension and anaemia on myocardial infarction and acute kidney injury. Specifically, we tested the primary hypothesis that a double hit of postoperative anaemia and hypotension synergistically worsens a 30-day composite of myocardial infarction and mortality. Secondarily, we tested the hypothesis that a double hit of postoperative anaemia and hypotension synergistically worsens in-hospital AKI.

Methods

This post hoc analysis of the POISE-2 trial (ClinicalTrials NCT01082874; PMID: 24679062) was approved by the Cleveland Clinic Institutional Review Board, Cleveland, Ohio, United States (IRB# 20–181, Executive Director: Bridget Howard) on 14 February 2020, with waived individual consent. The statistical analysis plan was developed a priori and registered with our Institutional Review Board before data were accessed for this analysis.

Briefly, the international POISE-2 trial enrolled 10 010 patients scheduled for major inpatient noncardiac surgery who had or were at risk for cardiovascular disease. Participants were factorially randomised to placebo or aspirin and to placebo or clonidine; and followed within first 30 postoperative days. The trial main results showed that neither aspirin nor clonidine reduced a composite of nonfatal myocardial infarction and all-cause mortality within 30 postoperative days. However, aspirin caused major bleeding and clonidine caused bradycardia and hypotension. There was no interaction between aspirin and clonidine.

Study population

All patients enrolled in the POISE-2 trial were eligible for this analysis. We excluded patients without available postoperative haemoglobin measurements or hypotension duration records, and those given only local anaesthesia.

Measurements

The exposures were lowest recorded haemoglobin concentration and hypotension defined by the average daily duration of SBP less than 90 mmHg. Both were considered between the end of surgery and the sooner of hospital discharge or postoperative day 4.

The primary outcome was a collapsed composite of nonfatal myocardial infarction and all-cause mortality during the initial 30 postoperative days. The diagnosis of myocardial infarction was based on the Third Universal Definition of Myocardial Infarction criteria, which was the most recent version available during the POISE-2 trial,[6] and verified by central adjudication. All-cause mortality was defined as any death during the initial 30 days after noncardiac surgery, regardless the cause.

Our secondary outcome was in-hospital AKI, based on all available postoperative creatinine concentrations while patients remained hospitalised or until postoperative day 7, whichever came first. Postoperative creatinine concentrations were compared with the last preoperative values within 30 days. We defined AKI according to the Kidney Disease Improving Clinical Outcomes (KDIGO) initiative published 2012 where serum creatinine increase to 1.5 to 1.9 times baseline within the last 7 days or a serum creatinine increase of at least $0.3\,g\,dl^{-1}$ from baseline within any postoperative 48 h as having AKI. As usual for perioperative studies, urine output was not considered (Supplemental Table 1, http://links.lww.com/EJA/A816).[7]

Statistical analyses

Due to potential overlap between exposure and outcome, we excluded patients whose lowest haemoglobin happened after the primary outcome event and only considered hypotension duration before the event. Patient characteristics, medical history, medication use and intraoperative management were summarised based on postoperative lowest haemoglobin groups: less than <8, ≥ 8– <11, ≥ 11– ≤ 13 and greater than $13\,g\,dl^{-1}$. However, continuous haemoglobin concentrations were used for analysis.

The primary outcome was a composite of nonfatal MI and all-cause mortality. We first fitted a multivariable Cox proportional hazard model with restricted natural splines of continuous lowest haemoglobin at 10^{th}, 50^{th} and 90^{th} percentiles to assess the linearity of the lowest haemoglobin concentration for the composite time-to-event outcome. We did not assess for linearity of hypotension duration because the range was small with 75% of patients having no exposure and the 0th to 90th percentile being 0 to 0.6 h per day.

We initially assessed the interaction between average daily hypotension duration and lowest haemoglobin spline term to test whether the relationship differed by hypotension duration, adjusting for confounders in Table 1. If the multiplicative interaction was significant, we would report the association between Hgb and the outcomes by hypotension status; if there were no significant multiplicative interaction, we would assess the marginal

Table 1 Characteristics of patients ($N = 7940$)

Factor	<8 g dl^{-1} ($N = 1037$)	8–11 g dl^{-1} ($N = 4249$)	11–13 g dl^{-1} ($N = 2163$)	>13 g dl^{-1} ($N = 491$)
Age	71 ± 10	71 ± 10	67 ± 10	65 ± 10
BMI	29 ± 7	30 ± 8	31 ± 7	30 ± 7
Female	599 (58)	2263 (53)	826 (38)	92 (19)
Race				
White Caucasian	822 (79)	3398 (80)	1714 (79)	338 (69)
Black/African descent	59 (6)	207 (5)	70 (3)	21 (4)
Hispanic/Latino	64 (6)	209 (5)	86 (4)	37 (8)
Asian	81 (8)	367 (9)	264 (12)	83 (17)
Other	12 (1)	68 (2)	29 (1)	12 (2)
Medical history				
Hypertension	891 (86)	3740 (88)	1878 (87)	399 (81)
Diabetes and taking insulin	366 (35)	1580 (37)	835 (39)	167 (34)
History of smoking within 2 years of surgery	211 (20)	951 (22)	667 (31)	209 (43)
History of coronary artery disease	203 (20)	845 (20)	480 (22)	160 (33)
History of peripheral vascular disease	86 (8)	362 (9)	190 (9)	54 (11)
History of stroke	69 (6.7)	195 (5)	96 (4)	40 (8)
History of transient ischemic attack	44 (4)	153 (4)	83 (4)	17 (4)
History of congestive heart failure	36 (4)	156 (4)	45 (2)	22 (5)
Baseline atrial fibrillation	27 (3)	107 (3)	54 (3)	15 (3)
Preoperative medication				
Beta blocker	312 (30)	1263 (30)	600 (28)	136 (28)
Statin	456 (44)	2115 (50)	1070 (50)	242 (49)
Aspirin	229 (22)	1014 (24)	487 (23)	132 (27)
Warfarin	33 (3.2)	81 (2)	35 (2)	10 (2)
Anticoagulant	182 (18)	597 (14)	365 (17)	77 (16)
Antiaggregant[a]	10 (1)	38 (1)	17 (1)	3 (0.6)
ACEI/ARB/direct renin inhibitor	581 (56)	2591 (61)	1294 (60)	240 (49)
Cox-2 inhibitor	66 (6)	345 (8)	121 (6)	14 (3)
NSAID/non-Cox-2 inhibitor	166 (16)	644 (15)	313 (15)	48 (10)
Rate-controlling calcium channel blocker	52 (5)	237 (6)	107 (5)	24 (5)
Dihydropyridine calcium channel blockers	241 (23)	1067 (25)	564 (26)	122 (25)
Insulin or oral diabetic drug	376 (36)	1624 (38)	852 (39)	174 (35)
Emergency/urgent surgery	81 (8)	220 (5)	146 (7)	29 (6)
Intraoperative bradycardia	259 (25)	1255 (30)	665 (31)	135 (28)
Anaesthesiology type				
General	430 (42)	1715 (40)	1120 (52)	304 (62)
Combined	317 (31)	1276 (30)	512 (24)	85 (17)
Epidural	15 (1)	75 (2)	31 (1)	12 (2)
Spinal	268 (26)	1142 (27)	490 (23)	87 (18)
Nerve block/plexus block	7 (1)	41 (1)	10 (1)	3 (1)
Surgery type				
Orthopaedic	518 (50)	2079 (49)	669 (31)	77 (16)
General	225 (22)	841 (20)	525 (24)	161 (33)
Urological	138 (13)	576 (14)	399 (18)	101 (20)
Vascular	55 (5)	268 (6)	172 (8)	37 (8)
Thoracic	36 (4)	239 (5)	206 (10)	46 (9)
Combined	31 (3)	99 (2)	49 (2)	10 (2)
Major spinal	21 (2)	74 (2)	56 (3)	19 (4)
Low risk	13 (1)	73 (2)	87 (4)	40 (8)
Surgery duration (h)	2.0 [1.0 to 4.0]	2.0 [1.0 to 3.0]	2.0 [1.0 to 3.0]	2.0 [1.0 to 2.0]
Surgical technique				
Open	904 (87)	3485 (82)	1465 (68)	274 (56)
Endoscopic	109 (11)	647 (15)	629 (29)	203 (41)
Both or other	24 (2)	117 (3)	69 (3)	14 (3)

Summary statistics are presented as mean ± SD, median [IQR] or *n* (%) as appropriately by postoperative haemoglobin concentration. All the variables listed above were adjusted for in all multivariable analyses. [a] Antiaggregant included thienopyridine and ADP receptor antagonist (nonthienopyridine).

relationship between each exposure and outcome while controlling for the other, as well as the potential confounding variables. Additionally, if we observed a nonlinear relationship between lowest haemoglobin and primary outcome, we would conduct a piece-wise linear Cox-proportional hazards model at a knot of 11 $g\,dl^{-1}$ of lowest haemoglobin, which enabled us to estimate hazard ratio differently when haemoglobin was below and above 11 $g\,dl^{-1}$.

We used similar methods to assess the relationship between hypotension and anaemia and in-hospital AKI. Two post hoc sensitivity analyses were conducted. For the first, we ignored outcome event time. Logistic regression models were fitted for both primary and secondary outcomes to assess the interaction between haemoglobin and hypotension duration and linearity.

Sample size

The POISE II database includes 10 010 patients, of whom 7% had a perioperative myocardial infarction or death. Based on a prior analysis, we anticipated that there would be 500 events among eligible patients. Assuming that the duration of SBP less than 90 mmHg is log-normally distributed with a mean of 1 min and a SD of 2.5 min, we expected to have more than 90% power at the 0.05 significance level to detect a clinically important hazard ratio as small as 1.05 per 5 min increase in minutes SBP less than 90 mmHg.

We also conducted a post hoc power calculation using simulation of a Cox-Proportional Hazards model with a follow-up time up 30 days and a censoring rate of 0.95. Assuming that main effect of haemoglobin was hazard ratio 0.95 for each unit increase, the main effect of hypotension was hazard ratio 1.40 (yes versus no), and the interaction effect was hazard ratio 0.95 per 1 unit increase in haemoglobin from 10 comparing with hypotension to without, we had 10% power to detect a significant interaction with our actual sample size of 7944. We were able to detect at least a hazard ratio of 0.78 for the interaction with power of 0.9 with the current sample size.

All analyses were performed in SAS Studio 3.7 (2012–2016 SAS Institute Inc., Cary, North Carolina, USA) or R studio Version 1.3.959 (2009–2020 RStudio, PBC).

Results

Ten thousand and ten patients from POISE-2 were eligible for the study and after various exclusions, 9990 patients remained. We further excluded 2050 patients because of missing postoperative haemoglobin measurements, missing confounders, and lowest haemoglobin recorded after the outcome event. Finally, 7940 patients were included in our analysis (Fig. 1). Among those excluded, 78% were missing postoperative haemoglobin, apparently because hospital stays were short. We, therefore, assumed the missingness was random, which means that missing data were related to some baseline data but not to outcomes and conducted a complete case analysis.[8]

The mean $\pm$ SD of lowest postoperative haemoglobin was $10 \pm 2\,g\,dl^{-1}$, and 24% of the patients had SBP less than 90 mmHg on any given day, with daily duration ranging from a minimum of 0 h to a max of 15 h per day (Supplemental Table 2, http://links.lww.com/EJA/A817). Four hundred nine (5.2%) patients had postoperative

Fig. 1 Flowchart.

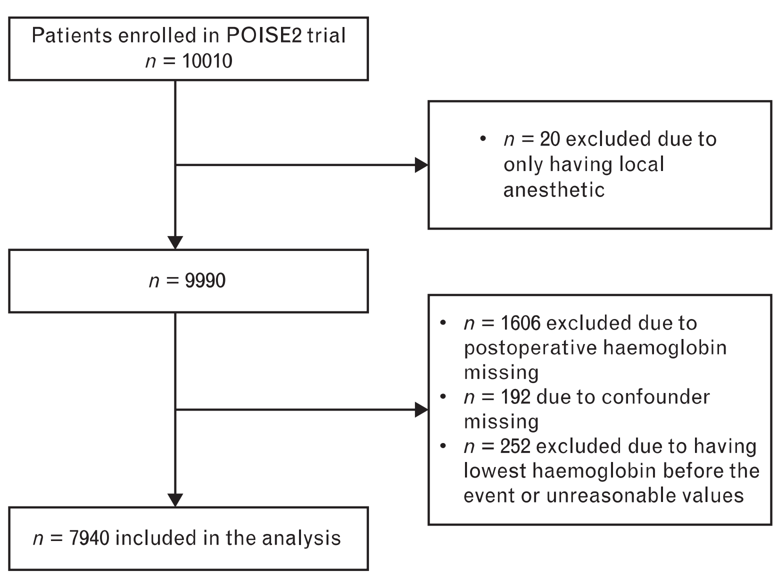


Table 2 **The association between both exposures and both outcomes**

Exposure	Estimated hazard ratio (95% CI)[a]	P values	Estimated hazard ratio (95% CI)[a]	P value
Main analysis	Primary outcome-MI/death ($n = 7940$)		Secondary outcome-postoperative AKI [d] ($n = 6523$)	
Interaction between haemoglobin spline and duration of hypotension per day	–	0.15	–	0.43
Minimum haemoglobin: main effect				
$<$11 g dl^{-1}, 1 unit decrease	1.30 (1.20 to 1.41)	$<$0.001	1.35 (1.25 to 1.46)	$<$0.001
$\geq$11 g dl^{-1}, 1 unit increase	0.94 (0.78 to 1.14)	0.54	1.28 (1.10 to 1.47)	$<$0.01
Duration of hypotension per day (hour): main effect	1.28 (1.20 to 1.36)	$<$0.001	1.22 (1.14 to 1.32)	$<$0.001
	Estimated odds ratio (95% CI)		Estimated odds ratio (95% CI)	
Sensitivity analysis	Primary outcome-MI/death ($n = 7940$)[b,c]		Secondary outcome-postoperative AKI ($n = 6523$)[b,c,d]	
Interaction between haemoglobin spline and duration of hypotension per day	–	0.26	–	0.61
Minimum haemoglobin: main effect				
$<$11 g dl^{-1}, 1 unit decrease	1.34 (1.23 to 1.46)	$<$0.001	1.43 (1.32 to 1.56)	$<$0.001
$\geq$11 g dl^{-1}, 1 unit increase	0.95 (0.78 to 1.15)	0.58	1.22 (1.05 to 1.42)	0.009
Duration of hypotension per day (hour): main effect	1.33 (1.23 to 1.46)	$<$0.001	1.25 (1.15 to 1.37)	$<$0.001

AKI, acute kidney injury; CI, confidence interval; Hgb, haemoglobin; MI: myocardial injury. [a] The main analysis was conducted through Cox-proportional hazard models adjusting for confounders in Table 1. Due to nonlinearity of the relationship between lowest haemoglobin and both outcomes, minimum haemoglobin was modelled as two-piece linear terms with a change point at 11 g dl^{-1}. The estimated hazard ratio is for each 1 g dl^{-1} decrease (Hgb $<$ 11) or each 1 g dl^{-1} increase (Hgb $\geq$ 11) in minimum haemoglobin (for haemoglobin exposure) or each hour increase in daily duration of SBP less than 90 mmHg (for hypotension duration exposure). [b] Sensitivity analysis ignoring outcome time was conducted through logistic regression model adjusting for confounders in the main analysis. [c] Estimated OR associated with each 1 g dl^{-1} decrease (Hgb $<$ 11) or each 1 g dl^{-1} increase (Hgb $\geq$ 11) in minimum haemoglobin (for haemoglobin exposure) or each hour increase in daily duration of SBP less than 90 mmHg (for hypotension duration exposure). [d] Six thousand five hundred and twenty-three patients were included in the secondary analysis because of some patients who did not have baseline or postoperative creatinine information and we excluded patients whose lowest Hgb happened after AKI from the study population.

composite outcome of myocardial infarction or death within 30 days after surgery, and 417 (6.4%) patients developed AKI during the initial 7 postoperative days.

Primary outcome

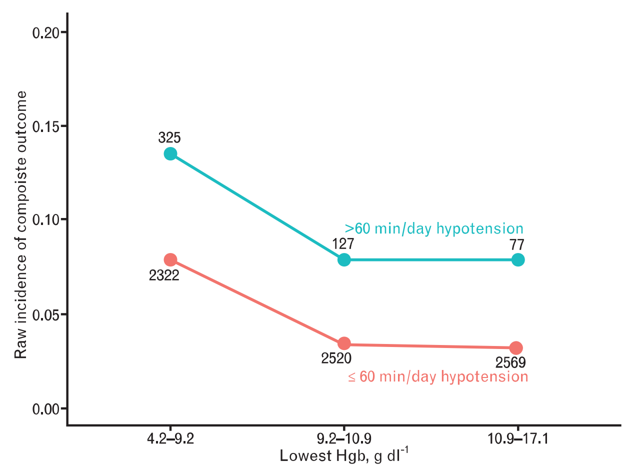
As reported previously,[9] there was a nonlinear relationship between lowest haemoglobin concentration and the composite outcome (spline $P < 0.001$), and a piece-wise linear Cox-proportional model with a knot at 11 g dl^{-1} of lowest haemoglobin was fitted (Supplemental Figure 1, http://links.lww.com/EJA/A815). Anaemia was associated with greater hazard for the composite outcome when the lowest haemoglobin level was below 11 g dl^{-1} (Table 2). As reported previously,[4] a duration of SBP less than 90 mmHg was associated with greater hazard for the primary composite outcome of nonfatal infarction and all-cause mortality (Table 2).

We did not find evidence of a significant multiplicative interaction between haemoglobin splines and hypotension duration on the association with the composite of nonfatal MI and all-cause mortality while treating hypotension duration either as continuous (average of hypotension duration in hours per day; $P = 0.15$), or dichotomously (having hypotension or not; $P = 0.96$, Fig. 2 and Supplemental Figure 1, http://links.lww.com/EJA/A815).

Secondary outcome

As reported previously,[9] lower lowest haemoglobin was associated with greater hazard for experiencing AKI when haemoglobin concentrations were less than 11 g dl^{-1} (spline term $P < 0.001$, Supplemental Figure 2, http://links.lww.com/EJA/A815). The duration of SBP less than 90 mmHg was also associated with higher hazard of AKI (Table 2). However, no multiplicative interaction was found between postoperative haemoglobin splines and hypotension on AKI while treating hypotension duration

Fig. 2 Raw incidence of composite outcome of myocardial injury and mortality based on haemoglobin and hypotension duration groups, univariate



Observed incidence of composite outcome of MI and mortality for each Hgb and hypotension duration group. The number next to the corresponding dot represents the sample size in each group. Hgb, haemoglobin; MI, myocardial injury; min, minute.

either continuously as the average duration of hypotension in hours per day ($P = 0.43$) or dichotomously ($P = 0.65$, Table 2, Fig. 3 and Supplemental Figure 2, http://links.lww.com/EJA/A815).

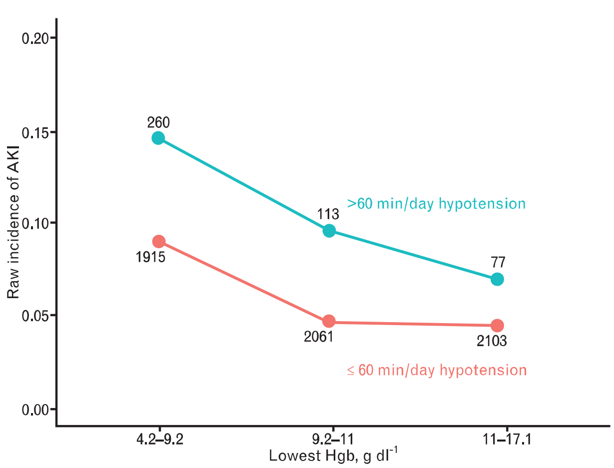
In sensitivity analyses, ignoring the timing of outcomes, we also did not find a synergistic effect of the lowest haemoglobin and hypotension duration on the association with the composite of nonfatal MI and all-cause mortality, or for AKI. The marginal (after dropping interaction term) relationship between both exposures and outcomes were similar to our primary and secondary analysis results (Table 2; Supplemental Figures 3–5, http://links.lww.com/EJA/A815).

Discussion

When two exposures are combined, they can influence outcomes in various ways. The most common is for the effects of each exposure to be additive. That is, for the combined effect of two exposures to equal the sum of the independent exposure effects on the outcome. From a statistical perspective, assuming adequate power, an additive interaction is indicated by lack of an interaction. Therefore, the lack of statistical interaction does not mean that the individual exposures have no effect on the outcome, only that the combined effect is additive. In contrast, a positive statistical interaction indicates a multiplicative (synergistic) relationship, and a negative interaction indicates that the exposures antagonistic each other.

Hypotension and anaemia were the two exposures we considered, and the primary outcome was a composite of myocardial infarction and mortality. As reported previously,[4,9] each exposure was distinctly associated with the composite outcome, and the magnitude of the associations were highly clinically meaningful. However, there was no significant interaction, indicating that the effects of hypotension and anaemia are most likely additive rather than multiplicative on the composite outcome of myocardial infarction and death.

Fig. 3 Raw incidence of acute kidney injury based on haemoglobin and hypotension duration groups, univariate



AKI, acute kidney injury; Hgb, haemoglobin; min: minute. Observed incidence of AKI in various Hgb and hypotension duration groups. The number next to the corresponding dot represents the sample size in each group.

There are two previous studies evaluating the interaction of hypotension and anaemia on AKI. In distinct contrast to our findings, Sickeler *et al.*[10] reported that hypotension did not worsen the renal effects of anaemia. Various factors may have contributed to differing outcomes. Sickeler *et al.*[10] evaluated patients having cardiac surgery, which is characterised by intense inflammation and cardiopulmonary bypass, both of which contribute to the AKI. Sickeler *et al.*[10] only considered the intraoperative period, especially cardiopulmonary bypass, whereas we included the postoperative period, which surely contributes to both myocardial infarctions and AKI. Furthermore, Sickeler *et al.*[10] included only cardiopulmonary bypass time and mean arterial pressures 50 mmHg or less whereas we considered duration of systolic pressure less than 90 mmHg. Haase *et al.*[11] also evaluated cardiac surgical patients, again only during cardiopulmonary bypass, and observed that anaemia superimposed on hypotension additively promoted renal injury – as did we.

A limitation of our analysis is that the POISE-2 trial recorded minutes of systolic pressure less than 90 mmHg during various phases of surgery and recovery, rather than actual pressures. We were thus unable to consider other hypotensive thresholds. We previously reported that typical vital sign assessments at 4 h intervals miss about half of all postoperative hypotension.[12] It is, therefore, likely that our patients actually experienced considerably more hypotension than was observed. Nonetheless, results were similar in a sensitivity analysis when we ignored the timing of outcomes, suggesting that denser exposure recording would not alter our overall conclusions.

The most serious limitation of our analysis is limited power for evaluating interactions. (Many more patients are required to evaluate interactions than the underlying relationships.) Nonetheless, our post hoc power calculation indicates that we had sufficient power to detect a hazard ratio of at least 0.78 for the interaction. Smaller interactions, thus, remain possible.

An additional limitation is that only systolic pressures were recorded in the underlying POISE-2 trial.[13,14] However, we have previously reported that systolic, mean and diastolic pressures are similarly associated with myocardial and renal injuries.[15] Similarly, only the lowest haemoglobin concentration and its date were recorded. We were, therefore, unable to consider haemoglobin changes over time. And finally, as in any observational analysis, unobserved confounding may distort our results.

However, the underlying trial was designed to evaluate the same primary outcome we considered and, therefore, including considerable detail about baseline risk and other potential confounding factors – allowing for careful statistical adjustment.

Conclusion

In summary, anaemia and hypotension are each strongly associated with the composite of myocardial infarction and mortality, and with AKI. However, there was no significant interaction between the exposures, thus suggesting that the interaction between them is additive rather than synergistic. Our results nonetheless suggest that it would be prudent to avoid the combination of anaemia and hypotension in patients recovering from major noncardiac surgery.

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Conflicts of interest: DIS is a consultant for Edwards Lifesciences, and Perceptive Medical. AT was a speaker for Pfizer and consultant for Consentric Medical. None of the conflicts are related to the submitted study.

Presentation: none

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